

Shivam Goel

goelshivam1210@gmail.com | [personal website](#) | [linkedin](#) | [github](#) | [twitter](#) | [scholar](#)

RESEARCH INTERESTS

- Reinforcement Learning
- Neurosymbolic Learning
- Object-Centric and Force-Based Robot Learning
- Task and Motion Planning (TAMP)
- Simulation-to-Real Transfer for Robotics
- Cognitive Robotic Architectures

EDUCATION

Tufts University

Ph.D. in Computer Science

Medford, MA

Jan 2020 – May 2026

- **Thesis:** Structured Abstractions for Learning and Adaptation in Open World Environments
- **Advisors:** [Prof. Jivko Sinapov](#) & [Prof. Matthias Scheutz](#)

Washington State University

Masters of Science in Computer Science

Pullman, WA

Aug 2015 – May 2017

Thesis: Engineering a Smart Scarecrow: Bird deterrence through UAS in blueberry fields

- **Advisor:** [Prof. Matthew E. Taylor](#)

Uttar Pradesh Technical University

Bachelors of Engineering in Information Technology

Noida, UP

Aug 2011 – June 2015

EXPERIENCE

Mulip Lab – Tufts University

Researcher · Advisor: Prof. Jivko Sinapov

Medford, MA

Jan 2020 – May 2026

- Designed and deployed RL policies for contact-rich, sustained object manipulation on real robotic platforms.
- Developed force-based, object-centric, robot-agnostic manipulation skills enabling generalization across robots and objects.
- Built integrated planning-and-learning frameworks for open-world robotic execution under novel objects and dynamics.
- Validated policies in simulation (MuJoCo, PyBullet) and deployed on physical robots.
- First-authored publications at ICRA-25, AAMAS-24, ICDL-22.

HRI Lab – Tufts University

Researcher · Advisor: Prof. Matthias Scheutz

Medford, MA

May 2023 – May 2026

- Developed neurosymbolic robotic systems integrating learned manipulation policies with task-and-motion planning.
- Designed robot-agnostic, force-based manipulation skills for non-prehensile and coordinated multi-robot interaction.
- Integrated learned policies into full robotic execution stacks for general-purpose task execution.
- Publications at AIJ, AAMAS-21, ICRA-25

IRL Lab – Washington State University

Researcher · Advisor: Prof. Matthew E. Taylor

Pullman, WA

Oct 2016 – Dec 2019

- Autonomous UAV system for bird detection and deterrence in agricultural environments using vision-based perception and closed-loop control (publications at ASABE).

- Robotic assistive systems for activities of daily living, integrating object detection and ROS-based deployment of custom turtlebot based robot in smart-home environments (Journal publication in Cognitive Systems Research).

Center of AI for Social Good – University of Southern California

Los Angeles, CA

Research Intern · Advisor: Prof. Milind Tambe & Prof. Bistra Dilkina

Apr 2018 – Aug 2018

- Applied AI to wildlife conservation, developing UAV-based thermal vision methods for poacher detection.
- Improved detection accuracy of Faster RCNN models for object detection.

ACTIVE PROJECTS

Generalizable Robotic Skills | *Physics-based Learning, Object-centric RL*

- Develop advanced motion primitives (e.g., twisting, non-prehensile object manipulation, planar pushing of heavy objects) to enable transferable robot skills across tasks and platforms.
- Paper under review at **IROS-26**

Scalable Frameworks for General-Purpose Robots | *Neurosymbolic AI, General-purpose Robotics*

- Leverage hybrid neurosymbolic architectures to integrate force-based robot-agnostic skills into larger robotic platforms capable of broad generalization.
- Journal in preparation to **Robotics Science & Systems**

Reasoning for Open-World Adaptation | *Novelty Handling, Affordances discovery, LLM-based PDDL-typing*

- Incorporate symbolic reasoning using LLMs to guide PDDL-typing for creative problem solving in Robots.
- Preparation for submission to **CoRL-2026**

PUBLICATIONS

- **Shivam Goel***, Shijie Fang*, Wenchang Gao*, Matthias Scheutz, Jivko Sinapov. [FLEX: A Framework for Learning Robot-Agnostic Force-based Skills Involving Sustained Contact Object Manipulation](#). **ICRA**, 2025.
- Pierrick Lorang, **Shivam Goel**, Yash Shukla, Patrick Zips, Matthias Scheutz. [A Framework for Neurosymbolic Goal-Conditioned Continual Learning for Open World Environments](#). **IROS**, 2024.
- James Staley, **Shivam Goel**, Yash Shukla, Elaine Schaefer Short. [Agent-Centric Human Demonstrations Train World Models](#). Reinforcement Learning Conference (**RLC**), 2024.
- **Shivam Goel**, Yichen Wei, Panagiotis Lympereopoulos, Klara Chura, Matthias Scheutz, Jivko Sinapov. [NovelGym: A Flexible Ecosystem for Hybrid Planning and Learning Agents Designed for Open Worlds](#). **AAMAS**, 2024.
- **Shivam Goel***, Panagiotis Lympereopoulos*, Ravenna Thielstrom, Evan Krause, Patrick Feeney, Pierrick Lorang, Sarah Schneider, Yichen Wei, Eric Kildebeck, Stephen Goss, Michael C. Hughes, Liping Liu, Jivko Sinapov, Matthias Scheutz. [A Neurosymbolic Cognitive Architecture Framework for Handling Novelty in Open Worlds](#). Artificial Intelligence Journal (**AIJ**), 2024.
- Yash Shukla, Bharat Kesari, **Shivam Goel**, Robert Wright, Jivko Sinapov. [Few-Shot Policy Transfer through Observation Mapping and Behavior Cloning](#). **IROS**, 2023.
- Pierrick Lorang, **Shivam Goel**, Patrik Zips, Jivko Sinapov, Matthias Scheutz. [Speeding-Up Continual Learning Through Information Gains in Novel Experiences](#). PRL Workshop @ **IJCAI**, 2022.
- **Shivam Goel***, Yash Shukla*, Vasanth Sarathy, Matthias Scheutz, Jivko Sinapov. [RAPid-Learn: A Framework for Learning to Recover for Handling Novelty in Open-World Environments](#). **ICDL**, 2022.
- **Shivam Goel***, Gyan Tatiya*, Matthias Scheutz, Jivko Sinapov. [NovelGridworlds: A Benchmark Environment for Detecting and Adapting to Novelty in Open Worlds](#). ALA Workshop @ **AAMAS**, 2021.
- Faizan Muhammad, Vasanth Sarathy, Gyan Tatiya, **Shivam Goel**, Saurav Gyawali, Mateo Guaman, Jivko Sinapov, Matthias Scheutz. [A Novelty-Centric Agent Architecture for Changing Worlds](#). **AAMAS**, 2021.

- Vasanth Sarathy, Daniel Kasenberg, **Shivam Goel**, Jivko Sinapov, Matthias Scheutz. [Spotter: Extending Symbolic Planning Operators Through Targeted Reinforcement Learning](#). **AAMAS**, 2021.
- Garrett Wilson, Christopher Pereyda, Nisha Raghunath, Gabriel de la Cruz, **Shivam Goel**, Sepehr Nesaei, Bryan Minor, Maureen Schmitter-Edgecombe, Matthew E. Taylor, Diane J. Cook. [Robot-Enabled Support of Daily Activities in Smart Home Environments](#). **Cognitive Systems Research**, 2019.
- Santosh Bhusal, Kapil Khanal, **Shivam Goel**, Manoj Karkee, Matthew E. Taylor. [Bird Deterrence in a Vineyard Using an Unmanned Aerial System](#). Transactions of the ASABE, 2019.
- **Shivam Goel**, Santosh Bhusal, Matthew E. Taylor, Manoj Karkee. [Detection and Localization of Birds for Bird Deterrence Using UAS](#). **ASABE Annual Meeting**, 2017.
- Santosh Bhusal, **Shivam Goel**, Kapil Khanal, Matthew E. Taylor, Manoj Karkee. [Bird Detection, Tracking and Counting in Wine Grapes](#). **ASABE Annual Meeting**, 2017.

* denotes equal contribution.

TEACHING EXPERIENCE

Tufts University

Medford, MA

Instructor — Introduction to Data Structures

May 2024 – Aug 2024

- Designed and delivered lectures covering core data structures and algorithms.
- Developed assignments and evaluated student performance.

Tufts University

Medford, MA

Teaching Assistant — Artificial Intelligence (Instructor: Prof. Fabrizio Santini)

Sep 2023 – May 2024

- Assisted students with AI theory and implementation of learning algorithms.
- Led class assistants for grading and assignment development

Washington State University

Pullman, WA

Teaching Assistant — Operating Systems (Instructor: Prof. K.C. Wang)

Sep 2018 – Dec 2019

- Co-developed programming assignments in C++ and assisted with grading.

Washington State University

Pullman, WA

Teaching Assistant — Software Engineering (Instructor: Prof. David Bakken)

Sep 2017 – May 2018

- Supported course projects and graded assignments in C#.

Washington State University

Pullman, WA

Teaching Assistant — Algorithms (Instructor: Prof. Zhe Dang)

Sep 2016 – Dec 2016

- Graded assignments and assisted students with algorithmic problem solving.

INVITED TALKS

Brown University

Aug 2025

Towards Building Autonomous AI Agents for Open-World Environments

Host: Prof. George Konidaris

Tufts University

Dec 2025

Reinforcement Learning for Force-Based Robot-Agnostic Policies

Host: Prof. Jivko Sinapov

TECHNICAL DEMONSTRATIONS

Tufts Robotics Open House | 2024, 2025

- Showcased neurosymbolic robotic learning systems on quadruped legged robots (Boston Dynamics Spot) and wheeled mobile manipulators (LoCoBot) for novelty handling to cross-disciplinary audience.
- Demonstrated Robotic System's ability to take voice commands, reason and act in the world.

MassRobotics | *Boston, 2023*

- Demonstrated VR-based interaction capabilities of a turtlebot

Tufts Graduate Showcase | *2022*

- Presented multimodal robotic interaction systems to cross-disciplinary audiences.

TECHNICAL EXPERTISE

Robotics: Contact-rich manipulation, force/torque learning, sim-to-real transfer, task-and-motion planning

Robots: UR5e, Kinova, Boston Dynamics Spot, LoCoBot (5-DOF), TurtleBot

Simulation: MuJoCo, PyBullet, Gazebo, Robosuite, Isaac Gym

Learning: PyTorch; RL/IL pipelines; OpenAI Gym-style environments

Middleware: ROS/ROS2

Programming: Python, C/C++, Java, MATLAB

MENTORING

- Yichen Wei. B.S., Tufts University. Undergraduate Researcher. (Currently M.S. student, Brown University.)
- Shijie Fang. M.S., Tufts University. Master's Researcher. (Currently Ph.D. student.)
- Klara Chura. B.S., Tufts University. Undergraduate Researcher. (Currently M.S. student, Brown University.)
- Turner Hayes. M.S., Tufts University. Master's Researcher. (Currently Front end Visualization Dev Boston Dynamics)
- Sam Sherman. B.S., Tufts University. Summer Intern.
- Tomas Maranga. B.S., Tufts University. Undergraduate Researcher.

SERVICE & OUTREACH

Professional Service

- PC Member: ICAPS-26, RLS-26, CoRL-26
- Reviewer: ICRA ('24-'26), IROS ('23-'26), AAMAS ('22, '24), JAIR, AIJ, IEEE RA+L, ICDL
- Workshop Co-organizer: 2nd IJCAI Workshop on Artificial Intelligence for Autonomous Driving (AI4AD), 2022.

Academic & Community Engagement

- Organizer and Coordinator: VLA Reading Group at Tufts (2025), RL Reading Group at Tufts SAIL-ON (2020), RL Reading Group at WSU (2019).
- Seminar Coordinator: IRL Lab bi-weekly seminar at WSU (2016-2017).
- Volunteer Photographer: Eastern Washington Aspirations in Computing Award Ceremony (2018-2019).

Last updated: May 2026